

KÉLIAN SOMMER

Postdoctoral Research Associate in astrophysics

 [kelian98.github.io](https://github.com/kelian98)

 +33 (0)7 77 91 83 62

 kelian.sommer@protonmail.com

 Dorlisheim, France

CURRENT POSITION

Jan. 2025 - present **Postdoctoral Research Associate** **University of Arizona, Tucson, USA**
Prof. Erika Hamden group at Steward Observatory in the College of Science
Development, optimization and testing of new, photon counting, scientific Skipper CCD detectors.
Work in a clean room environment with sensitive electronics.

RESEARCH INTERESTS

- **Instrumentation:** calibration of telescope instruments and sensors, instrumental and atmospheric transmissions, uncooled infrared thermal detectors, ultra-low noise skipper CCDs, command and control methods
- **Science:** CGM studies, type Ia supernovae and dark energy
- **Techniques:** photometry, radiometry, spectroscopy, data processing and analysis, noise modeling

EDUCATION AND TRAINING

Oct. 2021 - Nov. 2024 **PhD in Astrophysics** **Université de Montpellier, France**
Laboratoire Univers et Particules de Montpellier - CNRS/IN2P3
Subject: High-precision photometry in the era of ZTF and LSST cosmological surveys: characterization of instrumental and atmospheric transmissions
Graduated with honours from the jury
Advisors: Bertrand Plez, Johann Cohen-Tanugi, Eric Nuss

Sep. 2016 - Sep. 2021 **Master's degree in Aerospace Engineering** **ESTACA Engineering School, Paris-Saclay, France**
Propulsion systems, aerodynamics, operations
Orbital mechanics, Guidance Navigation & Control, mechanical, thermal and electrical architecture
Real-Time Operating Systems, microcontroller programming, sensors and actuators

2019 - 2021 **University diploma** **Observatoire de Paris, France**
Fundamentals for astrophysics (Graduate level)
Instrumentation, measuring chain and projects (Master level)

2019 - 2020 **Graduate level teaching units** **Faculté des Sciences et Ingénierie, Sorbonne Université, Paris, France**
Quantum mechanics
Electromagnetism and optics

PREVIOUS RESEARCH EXPERIENCE

Feb. 2021 - Aug. 2021 **Master's degree internship (2nd year)** **Université de Montpellier, France**
Laboratoire Univers et Particules de Montpellier - CNRS/IN2P3
Subject: Atmosphere characterization with an uncooled infrared thermal camera instrument prototype
Advisors: Eric Nuss, Bertrand Plez, Johann Cohen-Tanugi

Jun. 2020 - Sep. 2020 **Master's degree internship (1st year)** **Marseille, France**
Centre de Physique des Particules de Marseille - CNRS/IN2P3
Subject: Development of a control-command software for LISA space-based gravitational waves detector prototype bench
Advisor: Aurelia Secroun

TEACHING EXPERIENCE

Fall 2022 **Teaching assistant** **Université de Montpellier, France**
Supervision of observational astrophysics project at Observatoire de Haute-Provence for Master's students

PUBLICATIONS

[NASA ADS link](#) | [ORCID link](#)

Published

- M. Betoule, S. Antier, E. Bertin, P. É. Blanc, S. Bongard, J. Cohen Tanugi, S. Dagoret-Campagne, F. Feinstein, D. Hardin, C. Juramy, L. Le Guillou, A. Le Van Suu, M. Moniez, J. Neveu, É. Nuss, B. Plez, N. Regnault, E. Sepulveda, K. Sommer, T. Souverin, and X. F. Wang. StarDICE. I. Sensor calibration bench and absolute photometric calibration of a Sony IMX411 sensor. *A&A*, 670:A119, Feb. 2023
- B. Carreres, J. E. Bautista, F. Feinstein, D. Fouchez, B. Racine, M. Smith, M. Amenouche, M. Aubert, S. Dhawan, M. Ginolin, A. Goobar, P. Gris, L. Lacroix, E. Nuss, N. Regnault, M. Rigault, E. Robert, P. Rosnet, K. Sommer, R. Dekany, S. L. Groom, N. Sravan, F. J. Masci, and J. Purdum. Growth-rate measurement with type-Ia supernovae using ZTF survey simulations. *A&A*, 674:A197, June 2023
- T. Souverin, J. Neveu, M. Betoule, S. Bongard, S. Brownsberger, J. Cohen-Tanugi, S. Dagoret-Campagne, F. Feinstein, C. Juramy, L. Le Guillou, A. Le Van Suu, P. E. Blanc, F. Hazenberg, E. Nuss, B. Plez, E. Sepulveda, K. Sommer, C. Stubbs, N. Regnault, and E. Urbach. Measurement of telescope transmission using a Collimated Beam Projector. *arXiv e-prints*, page arXiv:2206.07530, June 2022
- K. Sommer, J. Cohen-Tanugi, B. Plez, M. Betoule, S. Bongard, L. Le Guillou, J. Neveu, E. Nuss, E. Sepulveda, T. Souverin, M. Moniez, and C. W. Stubbs. Design and performance of a collimated beam projector for telescope transmission measurement using a broad-band light source. *RAS Techniques and Instruments*, 3(1):125–142, 02 2024
- K. Sommer, B. Plez, J. Cohen-Tanugi, S. Dagoret-Campagne, M. Moniez, J. Neveu, M. Betoule, S. Bongard, F. Feinstein, L. Le Guillou, C. Juramy, E. Sepulveda, and T. Souverin. Stardice ii: Calibration of an uncooled infrared thermal camera for atmospheric gray extinction characterization. *Sensors*, 24(14), 2024
- K. Sommer, W. Kabalan, and R. Brunet. Infrared radiometric image classification and segmentation of cloud structures using a deep-learning framework from ground-based infrared thermal camera observations. *Atmospheric Measurement Techniques*, 18(9):2083–2101, 2025
- T. Souverin, J. Neveu, M. Betoule, S. Bongard, C. W. Stubbs, E. Urbach, S. Brownsberger, P. Blanc, J. Cohen-Tanugi, S. Dagoret-Campagne, F. Feinstein, D. Hardin, C. Juramy, L. Le Guillou, A. Le Van Suu, M. Moniez, Nuss, B. Plez, N. Regnault, E. Sepulveda, and K. Sommer. Stardice iii: characterization of the photometric instrument with a collimated beam projector. *RAS Techniques and Instruments*, 4:rza1010, 04 2025

In preparation

- K. Sommer, B. Plez, J. Cohen-Tanugi, M. Betoule, S. Bongard, S. Dagoret-Campagne, F. Feinstein, C. Juramy, L. Le Guillou, M. Moniez, J. Neveu, E. Sepulveda, and T. Souverin. StarDICE IV: characterizing atmosphere gray extinction with thermal infrared observations

TALKS

15 Dec. 2023	Rubin LSST-France meeting Recent progress of the long-wave infrared instrument for atmosphere monitoring within the StarDICE experiment	CC-IN2P3, Lyon, France
7 Dec. 2022	Journées Scientifiques et Techniques du Laboratoire Improving photometric calibration for type Ia supernovae cosmology in the era of wide-field surveys with the StarDICE experiment and the CBP instrument	LUPM, Montpellier, France
30 Nov. 2022	IN2P3 School of Instrumentation : "From detector to measurement" Systematic error reduction for type Ia supernovae cosmology with the photometric surveys ZTF and LSST	Centre CAES, Fréjus, France
13 May 2022	ZTF Spring Meeting Secondary maximum analysis for improved standardization of type Ia supernovae	LPNHE, Paris, France
26 Jan. 2022	56th Rencontres de Moriond 2022 - Cosmology StarDICE : instrumental flux calibration with an artificial star for type Ia supernovae cosmology with the Legacy Survey of Space and Time	La Thuile, Italy

COLLABORATIONS

Oct. 2021 - present	The Dark Energy Science Collaboration of the Vera Rubin Observatory Legacy Survey of Space and Time Involvement in the Photometric Calibration Working Group
Feb. 2021 - present	The StarDICE collaboration Metrology experiment designed to measure reference spectrophotometric CALSPEC stars to the mmag level relative to laboratory flux for type Ia supernovae cosmology
Oct. 2021 - Oct. 2024	Zwicky Transient Facility Technical contribution to the development of an instrument capable of measuring the telescope system's throughput to the mmag level

TECHNICAL AND COMPUTATIONAL SKILLS

- **Operating systems/environments:** UNIX/Linux, Windows, Docker
- **Computer languages:** Python, C/C++, Bash, $\LaTeX$, SQL
- **Astronomical software:** SourceExtractor, SaoDS9, PixInsight
- **CAD:** Catia, Solidworks, FreeCAD, Fusion360
- **Miscellaneous/tools:** 3D printing, soldering, Git

OUTREACH

2022 - present	Société Française d'Astronomie et d'Astrophysique (SF2A)	Association member
2022 - 2024	Société Astronomique de France (SAF)	Association member
2022 - 2023	Maison des Jeunes et de la Culture Astronomy club manager	Castelnau-le-Lez, France

LANGUES

French – native

English – fluent

German – basic

Spanish – basic

HOBBIES AND INTERESTS

Sports : running (marathon and trail), badminton, road cycling

Hobbies : hiking, amateur astronomy and astrophotography, night photography